



FACULTAD DE INGENIERÍA
Operations Research

Optimization of Domiciliary Waste Collection Using Vehicle Routing on Real Data from the Intendencia de Montevideo

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1. INTRODUCTION

Urban waste management is one of the main operational challenges faced by modern cities. In particular, the planning of collection routes has a direct impact on the operational efficiency of municipal services: an inefficient route implies greater fuel consumption, longer operating times and, consequently, higher costs for the system.

Historically, domiciliary waste collection in Montevideo relied on a bag-on-curb system, where citizens left their waste out on designated days and times, and collection trucks had to drive along every street in the circuit. Under that scheme, route planning can be represented as a Chinese Postman Problem, since the truck must cover every arc of the street network at least once. However, the Intendencia de Montevideo, through its División Limpieza, has migrated to a system of georeferenced domiciliary containers organised into collection circuits. This change transforms the nature of the problem: because each container has a fixed and known location, the truck no longer needs to cover every street, but rather visit the specific points where the containers are located.

Therefore, the objective of this project is to apply Operations Research tools to determine an efficient route for a waste-collection truck to visit all domiciliary containers in a selected circuit, minimising the total distance travelled. The model is formulated as an integer linear programming problem and solved with the Gurobi optimiser. In the implemented version, a single truck is considered, starting and ending at the Usina Felipe Cardoso, with a maximum shift duration and a capacity limit.

To better represent the urban road network, the model does not rely only on Euclidean distances between containers. Instead, the coordinates of the containers are used to compute road-network distances through OSRM. In addition, to reduce computational complexity, the set of possible arcs is restricted to connections between nearby nodes, while preserving all connections from and to the depot. This produces a simplified but operationally coherent routing model.

The model is based on the approach of Wesner (2015), who applies similar optimisation techniques to waste-collection routing in the Municipality of Morón, adapting the general methodology to the context of Montevideo and its georeferenced container system.

The data used come from the Open Data Portal of the Intendencia de Montevideo, published by the División Limpieza, and include the location of each container, its assigned collection circuit and its scheduled collection frequency.

2. PROBLEM DESCRIPTION

The problem consists of determining the shortest route for a single waste-collection truck that departs from the Usina Felipe Cardoso, the city's final

disposal site, visits every domiciliary container in a given circuit exactly once, and returns to the same starting point. The truck has a maximum load capacity of 8,000 kg and a shift limit of 480 minutes. Since the original dataset does not include the amount of waste generated by each container, a fixed demand of 2.55 kg per container is assumed for modeling purposes. In addition, a fixed service time of 1.5 minutes is assumed for the collection operation at each container.

The problem is formulated as a single-vehicle capacitated routing problem with a time constraint and solved using integer linear programming. To better represent the actual movement of the truck, distances between nodes are computed over the road network using OSRM, based on the georeferenced coordinates of the containers and the depot. Travel times are then estimated from these road-network distances using an average vehicle speed of 25 km/h. In order to reduce computational complexity, the set of feasible arcs is restricted to connections between nearby nodes, considering the 10 nearest neighbours of each node according to Euclidean distance, while maintaining all connections between the depot and the containers.

3. MODEL FORMULATION

Sets

- $N = \{0, 1, \dots, n\}$: Set of all nodes. Node 0 is the depot (Usina Felipe Cardoso); nodes 1 to n are the domiciliary containers.
- $C = N \setminus \{0\}$: Set of containers.
- $A = \{(i, j) \in N \times N : i \neq j\}$: Set of feasible directed arcs. Instead of allowing every pair of nodes to be connected, each node is connected only to its r nearest neighbours, with $r = 10$ and $i \neq j$. Additionally, all arcs from the depot to each container and from each container to the depot are included.

Parameters

- d_{ij} : Road-network distance between nodes (i) and (j), in metres, computed using OSRM from the georeferenced coordinates of the depot and containers. This parameter is defined only for feasible candidate arcs $((i,j)$ in A).
- $q_i = 2.55$ kg: Waste demand of container i , for all $i \in C$.
- $Q = 8,000$ kg: Maximum load capacity of each truck.
- $T_{\max} = 480$ min: Maximum shift duration (morning shift 06:00–14:00).
- $s_i = 1.5$ min: Fixed service time per container. Since real collection times are unavailable in the dataset, a fixed average is assumed.
- t_{ij} : Estimated travel time in minutes, computed as $(d_{ij} / 1000) / 25 \times 60$, assuming an average truck speed of 25 km/h.

Decision Variables

- $x_{ij} \in \{0, 1\}$: Equals 1 if the truck travels directly from node i to node j , and 0 otherwise, for all feasible candidate arcs.
- $u_i \geq 0$: Cumulative load of the truck upon arriving at container (i), used as an MTZ auxiliary variable for subtour elimination and capacity control, for all $i \in C$.

Objective Function

The objective is to minimise the total road-network distance travelled by the truck. No cost per kilometre is incorporated, since no reliable cost data are available; therefore, the focus is on operational efficiency.

$$\text{Min } Z = \sum_{(i,j) \in A} d_{ij} \cdot x_{ij}$$

This objective function adds the distance of each selected arc in the route. Since $x_{ij} = 1$ only when the truck travels directly from node i to node j , only the arcs used in the final route contribute to the total distance.

Constraints

(1) Incoming arc to each container

Each container must have exactly one incoming arc:

$$\sum_{i:(i,j) \in A} x_{ij} = 1, \quad \forall j \in C$$

This constraint ensures that every container is reached exactly once by the truck, preventing missed collections and duplicate visits. In other words, for each container j , exactly one selected arc must enter that container.

(2) Outgoing arc from each container.

Each container must have exactly one outgoing arc:

$$\sum_{j:(i,j) \in A} x_{ij} = 1, \quad \forall i \in C$$

This constraint ensures that the truck leaves each container exactly once after visiting it. Together with the incoming-arc constraint, it guarantees that every container is included exactly once in the route, preventing both missed collections and duplicate visits.

(3) Departure from and return to the depot:

$$\sum_{j \in C} x_{0j} = 1$$

$$\sum_{i \in C} x_{i0} = 1$$

Node 0 represents the Usina Felipe Cardoso, defined as the depot and final destination of the route. The first constraint ensures that the truck leaves the depot exactly once, while the second ensures that it returns to the depot exactly once, forming a closed route.

(4) Subtour elimination (MTZ):

$$u_i + q_j \leq u_j + Q(1 - x_{ij}), \quad \forall i, j \in C, i \neq j, (i,j) \in A$$

These MTZ constraints prevent subtours, that is, disconnected cycles among containers that do not include the depot. If the truck travels from container i to container j , the cumulative load upon arriving at j must be at least the cumulative load at i plus the demand of container j . This creates a consistent visiting order and helps eliminate disconnected cycles.

(5) Load bounds:

$$q_i \leq u_i \leq Q, \quad \forall i \in C$$

These constraints ensure that the cumulative load variable remains within feasible bounds. Upon arriving at each container, the accumulated load must be at least the demand of that container and cannot exceed the truck capacity $Q=8,000$ kg. Since a fixed demand of 2.55 kg per container is assumed, the capacity limit is not expected to be binding for the selected pilot circuit; the shift duration constraint is more restrictive.

(6) Maximum shift duration:

$$\sum_{(i,j) \in A} t_{ij} \cdot x_{ij} + \sum_{i \in C} s_i \leq T_{\max}$$

The shift lasts 480 minutes (06:00–14:00). The truck's total operating time is the sum of the travel time between selected nodes and the service time at each container. Road-network distances are computed using OSRM, and travel times are estimated from these distances assuming an average speed of 25 km/h. A fixed service time of 1.5 minutes per container is added because real collection times are not recorded in the dataset.

(7) Variable domains:

$$x_{ij} \in \{0, 1\} \quad u_i \geq 0$$

The routing variables are binary, meaning that each feasible arc is either selected or not selected in the final route. The cumulative load variables are continuous and non-negative. Fractional routing decisions, such as using half of an arc, are not operationally meaningful.

Summary of modeling decisions

The model was formulated as a single-vehicle Capacitated Vehicle Routing Problem with a time constraint. The main decisions were: defining the Usina Felipe Cardoso as the origin and destination of the route; assuming a fixed demand of 2.55 kg per container due to the absence of real demand data; setting the truck capacity at 8,000 kg; limiting the route duration to a maximum of 480 minutes; computing road-network distances with OSRM based on the georeferenced coordinates of the depot and containers; estimating travel times from these distances using an assumed average speed of 25 km/h; restricting feasible arcs to the 10 nearest neighbours of each node, while preserving all depot-container connections; and assuming a fixed service time of 1.5 minutes per container. These decisions produce a simplified but operationally coherent model, consistent with the available data and computational limitations.

4. COMPUTATIONAL RESULTS

The model was solved using Gurobi. The final formulation contained 1.109 constraints, 1.027 variables and 5.372 nonzero coefficients. Of the decision variables, 948 were binary routing variables associated with the feasible candidate arcs, while 79 were continuous cumulative load variables used in the MTZ constraints.

Gurobi found an optimal solution in 20.11 seconds, exploring 52.536 branch-and-bound nodes. The final objective value was 48.802,80 metres, equivalent to a total travelled distance of 48,80 km. The optimality gap was 0.0000%, meaning that the best feasible solution matched the best bound found by the solver.

The resulting route starts and ends at the Usina Felipe Cardoso and visits all 79 containers exactly once. The estimated travel time was 117.13 minutes. Since a fixed service time of 1.5 minutes was assumed for each of the 79 containers, the total service time was 118.50 minutes. Therefore, the total estimated route duration was 235.63 minutes, which is below the maximum shift duration of 480 minutes.

In operational terms, the obtained route uses approximately 49.1% of the available shift time:

$$\frac{235,63}{480} \times 100 \approx 49,1\%$$

Thus, the time constraint is satisfied with a remaining slack of approximately 244.37 minutes. This indicates that, under the assumptions of the model, the selected circuit can be served by a single truck within one shift.

5. SENSITIVITY ANALYSIS

1. Base scenario.

The base scenario corresponds to the parameter values used in the main optimization model. The truck is assumed to travel at an average speed of 25 km/h, each container requires 1.5 minutes of service time, and the feasible arc set is built using the 10 nearest neighbours of each node. Under these assumptions, Gurobi found an optimal route with a total road-network distance of 48.80 km. The estimated travel time was 117.13 minutes, while the total service time was 118.50 minutes. Therefore, the complete route duration was 235.63 minutes, which is well below the maximum shift duration of 480 minutes.

Parameter	Base value
Number of containers	79
Truck capacity (Q)	8,000 kg
Demand per container (q_i)	2.55 kg
Maximum shift duration ($T_{\max}$)	480 min
Average speed (v)	25 km/h
Service time (s_i)	1.5 min
Number of nearest neighbours (K)	10

Table 1: Base scenario parameter values.

Result	Base scenario
Total distance	48.80 km
Travel time	117.13 min
Service time	118.50 min
Total route duration	235.63 min
Remaining shift time	244.37 min
Solver time	20.11 s
Optimality gap	0.00%

Table 2: Base scenario computational results.

The base solution uses approximately 49.1% of the available shift time, leaving 244.37 minutes of slack. This suggests that, under the base assumptions, the selected circuit can be completed comfortably by a single truck within one shift.

2. Variation of average speed.

The average travel speed was varied in order to evaluate the effect of traffic conditions on the feasibility of the route. The base scenario assumes an average speed of 25 km/h. Since the objective function minimises total distance and not total travel time, the route obtained in the base scenario was kept fixed for this sensitivity test. Travel times were then recalculated for different average speeds in order to assess

whether the route remains feasible under slower or faster traffic conditions. The travel time was computed as:

$$T_{\text{travel}}(v) = \frac{D}{v} \times 60$$

Where the total distance is 48.80 km and v is the assumed average speed in km/h. Since the road-network distance remains fixed, changing the speed only affects the estimated travel time, while the service time remains constant at 118.50 minutes.

Average speed (v)	Travel time	Service time	Total route duration	Remaining shift time	Feasible?
15 km/h	195.21 min	118.50 min	313.71 min	166.29 min	Yes
20 km/h	146.41 min	118.50 min	264.91 min	215.09 min	Yes
25 km/h	117.13 min	118.50 min	235.63 min	244.37 min	Yes
30 km/h	97.61 min	118.50 min	216.11 min	263.89 min	Yes
35 km/h	83.66 min	118.50 min	202.16 min	277.84 min	Yes

As expected, lower speeds increase the total route duration, while higher speeds reduce it. However, even under the most conservative scenario considered, with an average speed of 15 km/h, the total estimated duration is 313.71 minutes. This remains below the maximum shift duration of 480 minutes. Therefore, the route remains feasible for all tested speed values.

This indicates that the solution is not highly sensitive to moderate changes in the average speed. The shift duration constraint remains non-binding in all tested scenarios, with at least 166.29 minutes of slack available even at 15 km/h.

3. Variation of service time.

The service time per container was varied in order to evaluate the effect of longer or shorter collection operations on the feasibility of the route. The base scenario assumes a fixed service time of 1.5 minutes per container. Since the objective function minimises total distance and not service time, the route obtained in the base scenario was kept fixed for this sensitivity test. The total route duration was then recalculated for different service time values.

The total route duration was computed as:

$$T_{\text{total}}(s) = T_{\text{travel}} + |C|s$$

Using the base route, this becomes:

$$T_{\text{total}}(s) = 117.13 + 79s$$

Where s is the assumed service time per container, in minutes.

Service time per container (s)	Travel time	Service time	Total route duration	Remaining shift time	Feasible?
0.5 min	117.13 min	39.50 min	156.63 min	323.37 min	Yes
1.0 min	117.13 min	79.00 min	196.13 min	283.87 min	Yes
1.5 min	117.13 min	118.50 min	235.63 min	244.37 min	Yes
2.0 min	117.13 min	158.00 min	275.13 min	204.87 min	Yes
3.0 min	117.13 min	237.00 min	354.13 min	125.87 min	Yes
4.0 min	117.13 min	316.00 min	433.13 min	46.87 min	Yes

As expected, increasing the service time per container increases the total route duration linearly. However, all tested values remain feasible within the 480-minute shift limit. Even when the service time is increased to 4.0 minutes per container, the total route duration is 433.13 minutes, leaving 46.87 minutes of slack.

This suggests that the solution is more sensitive to service time than to average speed, since service time is accumulated at every container. Nevertheless, under the tested scenarios, the selected circuit can still be completed by a single truck within one shift.

4. Computational Results.

To validate the operational feasibility of the obtained solution, the optimal routing sequence was projected onto the real street network of Montevideo. A dedicated Python script was developed utilizing the Folium library for interactive

cartographic rendering. Furthermore, to accurately represent the physical displacement of the collection truck, the Open Source Routing Machine (OSRM) API was integrated into the visualization pipeline. By converting the nodes' georeferenced coordinates and querying the OSRM engine, the generated map traces the exact driving path of the vehicle. This approach ensures that the simulated route strictly adheres to the city's actual road infrastructure, respecting traffic directions, valid turns, and street connectivity, thereby confirming the practical applicability of the theoretical mathematical model.

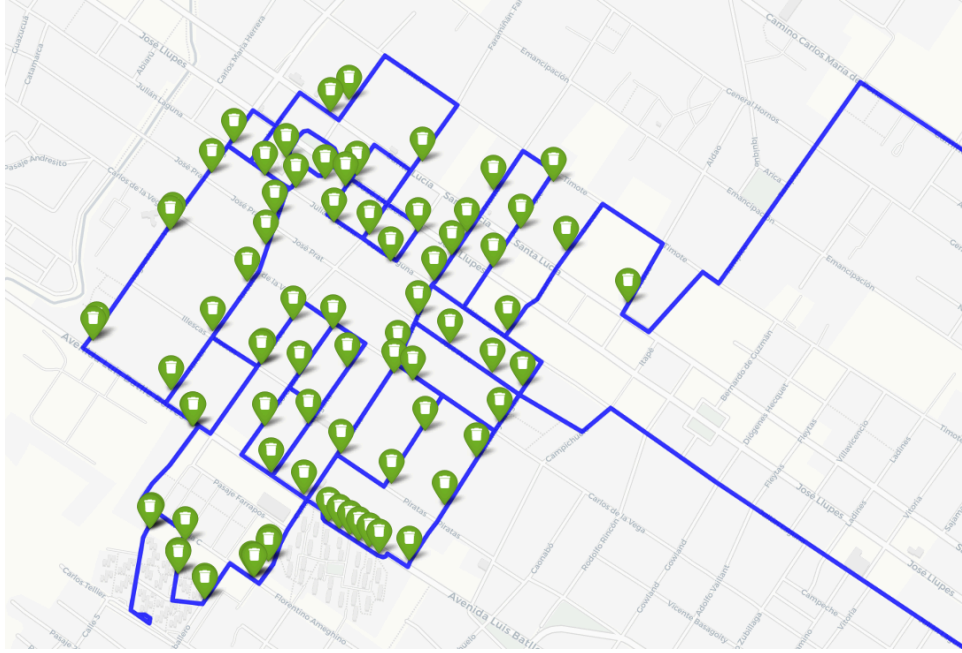


Figure 1: Map visualization of the optimal collection route for circuit A_DU_RM_CL_110, plotted over the road network using Folium and OSRM.

5. Variation of the number of nearest neighbours.

The number of nearest neighbours K was varied in order to evaluate the effect of the candidate arc selection strategy on solution quality and computational performance. Unlike the previous sensitivity tests, changing K modifies the feasible arc set A , and therefore the optimization model must be solved again for each value of K .

For each node, candidate arcs are generated towards its K nearest neighbours according to Euclidean distance. In addition, all arcs from the depot to each container and from each container back to the depot are preserved, in order to ensure that the truck can always leave from and return to the Usina Felipe Cardoso.

$$A_K = \{(i, j) : j \in NN_K(i), i \in N, j \in N, i \neq j\} \cup \{(0, j), (j, 0) : j \in C\}$$

Where $NN_K(i)$ denotes the set of the K nearest neighbours of node i .

(K) nearest neighbours	Candidate arcs	Solution status	Total distance	Total route duration	Solver time	Optimality gap
5	553	No feasible solution found within time limit	—	—	600.02 s	—
8	790	Optimal	48.87 km	235.78 min	59.19 s	0.0000%
10	948	Optimal	48.80 km	235.63 min	20.11 s	0.0000%
12	1,106	Optimal	48.80 km	235.63 min	96.23 s	0.0000%
15	1,343	Optimal within tolerance	48.80 km	235.63 min	58.83 s	0.0097%

For $K=5$, the model contained 553 binary routing variables and 632 total variables. However, no feasible integer solution was found within the 600-second time limit. This does not necessarily prove that the instance is infeasible for $K=5$, but it indicates that the candidate arc set is too restrictive for practical use in this case. With only five nearest neighbours per node, the model may not provide enough connectivity to construct a complete route visiting all containers and returning to the depot.

For $K=8$, the model generated 790 candidate arcs and found an optimal solution within 59,19 seconds. The resulting distance was 48,87 km, only 0,07 km longer than the base scenario with $K=10$. This indicates that $K=8$ provides enough connectivity to construct a feasible and high-quality route. However, despite having fewer binary variables, it required more computational time than $K=10$, suggesting that a more restrictive arc set can make the search process harder by reducing routing flexibility.

For $K=12$, the model generated 1.106 candidate arcs and found an optimal solution in 96,23 seconds. The total distance was 48,80 km, equal to the base case with $K=10$. This indicates that the additional arcs introduced by increasing K from 10

to 12 did not improve the objective value. Instead, they increased the model size and computational effort. Therefore, $K=12$ provides no operational advantage over $K=10$ for this instance.

For $K=15$, the model generated 1.343 candidate arcs and found a solution with the same total distance as the base case, equal to 48,80 km. The model was solved within Gurobi's optimality tolerance, with a final gap of 0.0097%. Although the solution quality was equivalent to the one obtained with ($K=10$), the model size increased considerably. Therefore, increasing K to 15 did not improve the route distance and added unnecessary computational complexity.

The results show that $K=1$ provides the best balance between solution quality and computational performance. With $K=5$, the arc set was too restrictive and no feasible solution was found within the time limit. Increasing K to 8 allowed the model to find an optimal solution, but with a slightly longer route than the base case. For $K=10$, $K=12$, and $K=15$, the best route distance was the same, equal to 48,80 km. However, $K=10$ required the lowest computational time among these three cases. Therefore, the base value $K=10$ was retained as the preferred configuration.

Although larger values of K increase the number of available arcs and provide more routing flexibility, they also increase the number of binary variables and the size of the branch-and-bound search. In this instance, adding arcs beyond $K=10$ did not improve the objective value.

6. Variation of truck capacity.

The truck capacity Q was also analysed in order to determine whether the capacity constraint is binding in the selected circuit. In the base scenario, the truck capacity is set to 8,000 kg. Since the model assumes a fixed demand of 2.55 kg per container and the circuit contains 79 containers, the total estimated demand is:

$$Q_{\min} = \sum_{i \in C} q_i = |C| \cdot q_i = 79 \cdot 2.55 = 201.45 \text{ kg}$$

This means that the total estimated waste demand of the circuit is only 201.45 kg, which is far below the truck capacity of 8,000 kg. Therefore, the capacity constraint is not binding in the base scenario.

The route remains feasible for any capacity value greater than or equal to 201.45 kg. For capacity values below this threshold, the model would become infeasible, because a single truck would not be able to collect the total demand of all

containers in the circuit. As a result, variations in Q above this minimum value do not affect the optimal route distance or the total route duration.

This confirms that, for the selected pilot circuit, the operational feasibility of the route is driven mainly by the time constraint and the candidate arc structure, rather than by truck capacity.

7. CONCLUSIONS

This project successfully demonstrated the application of Operations Research tools to optimize domiciliary waste collection in Montevideo. By formulating a single-vehicle Capacitated Vehicle Routing Problem (CVRP) and solving it using Gurobi, an optimal route of 48.80 km was found, ensuring all 79 containers in the selected circuit are visited exactly once before returning to the Usina Felipe Cardoso depot.

The modeling strategy proved to be highly efficient. Restricting the feasible arcs to the 10 nearest neighbours ($K=10$) provided the best balance between solution quality and computational performance, achieving a 0.0000% optimality gap in just 20.11 seconds.

Operationally, the computational results confirm that a single truck can comfortably serve the circuit within one shift. The estimated total route duration is 235.63 minutes, which utilizes only 49.1% of the 480-minute maximum shift, leaving a significant slack of 244.37 minutes. Furthermore, the sensitivity analysis validated the robustness of this solution. The route remains fully feasible even under adverse operational conditions, such as reducing the average travel speed to 15 km/h or increasing the collection service time to 4.0 minutes per container.

Finally, the analysis confirmed that the 8,000 kg truck capacity constraint is not binding. With a total estimated circuit demand of only 201.45 kg, the operational feasibility of the route is primarily driven by the time constraint and the candidate arc structure, rather than by the physical load limits of the vehicle.

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APPENDICES

```

Explored 52536 nodes (1134911 simplex iterations) in 20.77 seconds (15.95 work units)
Thread count was 16 (of 16 available processors)

Solution count 10: 48802.8 48802.8 48812.3 ... 49087.3

Optimal solution found (tolerance 1.00e-04)
Best objective 4.880280000000e+04, best bound 4.880280000000e+04, gap 0.0000%

=====
¡Ruta Óptima (o mejor encontrada) finalizada!
Distancia Total Recorrida: 48802.80 metros
Distancia Total Recorrida: 48.80 km
=====

Tiempo de viaje: 117.13 minutos
Tiempo de recolección: 118.50 minutos
Tiempo total estimado: 235.63 minutos

Arcos usados en la solución:
[(0, 39), (1, 3), (2, 5), (3, 19), (4, 41), (5, 78), (6, 70), (7, 35), (8, 43), (9, 14), (10, 75), (11, 23), (12, 40), (13, 22), (14, 62), (15, 8), (16, 47), (17, 16), (18, 57), (19, 45), (20, 37), (21, 71), (22, 63), (23, 69), (24, 17), (25, 66), (26, 4), (27, 73), (28, 20), (29, 61), (30, 58), (31, 72), (32, 38), (33, 29), (34, 6), (35, 12), (36, 77), (37, 60), (38, 64), (39, 74), (40, 55), (41, 76), (42, 9), (43, 28), (44, 7), (45, 59), (46, 1), (47, 36), (48, 32), (49, 26), (50, 46), (51, 31), (52, 27), (53, 79), (54, 50), (55, 11), (56, 68), (57, 54), (58, 67), (59, 65), (60, 10), (61, 0), (62, 18), (63, 53), (64, 34), (65, 49), (66, 2), (67, 13), (68, 44), (69, 51), (70, 52), (71, 25), (72, 42), (73, 15), (74, 24), (75, 30), (76, 48), (77, 56), (78, 33), (79, 21)]

Secuencia del Camión (Nodos):
[0, 39, 74, 24, 17, 16, 47, 36, 77, 56, 68, 44, 7, 35, 12, 40, 55, 11, 23, 69, 51, 31, 72, 42, 9, 14, 62, 18, 57, 54, 50, 46, 1, 3, 19, 45, 59, 65, 49, 26, 4, 41, 76, 48, 32, 38, 64, 34, 6, 70, 52, 27, 73, 15, 8, 43, 28, 20, 37, 60, 10, 75, 30, 58, 67, 13, 22, 63, 53, 79, 21, 71, 25, 66, 2, 5, 78, 33, 29, 61, 0]

El camión inicia en el Nodo 0 (Usina), visita 79 contenedores y retorna a la Usina.

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Figure 2: Solver output for the base scenario.